

Editors

IEEE Access

Dear Editors,

Please consider our manuscript entitled

“When Agreement Fails: Visual Anchoring and Multimodal Pseudo-Label Reliability in Classroom Behavior Recognition”

for publication in *IEEE Access*.

We submit this paper as a reliability-audit study of multimodal large language model (MLLM) pseudo-labeling in classroom behavior recognition. Using three SCB sub-datasets, we organise the evidence into four linked analysis modules: a protocol-matched zero-shot CLIP baseline, MLLM annotation and cross-model validation, CLIP adaptation under label-source choices, and mechanism/strategy audits. The study addresses a practical deployment question that is increasingly important across computer vision and educational AI: when can general-purpose MLLMs provide useful pseudo-label supervision, and when does cross-model agreement simply preserve shared error?

The manuscript makes four main contributions. First, it provides a reproducible audit protocol that links annotation quality, downstream adaptation, retention controls, distribution-shift checks, anchor-proxy analysis, auxiliary filtering and self-training baselines, and cross-model robustness. Second, it shows that pseudo-label usefulness depends on category-level visual evidence. Third, it demonstrates that agreement filtering is not a portable denoising rule: in the observed setting, agreement-filtered subsets can be affected by data-volume loss, class-composition shift, and shared label error. Fourth, it separates these boundary conditions from a single annotator choice or optimisation path using repeated-seed downstream evaluation, targeted LoRA replications, train-split teacher–student and confidence-filtering baselines, and a five-model robustness check.

We believe the manuscript is well suited to *IEEE Access* because it offers a reproducible, application-driven study at the intersection of multimodal AI, computer vision, and intelligent education. The emphasis is not another benchmark comparison; it is an auditable methodology and evidence base for designing reliable MLLM-based annotation workflows in classroom sensing and related vision systems.

This submission is original, has not been published previously, and is not under consideration elsewhere. All authors have approved the submitted version and agree to its submission to *IEEE Access*. The authors declare no competing interests.

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Data and code availability: The SCB dataset is publicly available at <https://huggingface.co/datasets/wintonYF/SCB-Dataset>. The experiment code is available at <https://github.com/zhanglizhuo/llm-annotation>.

Ethics and privacy: This study is a secondary analysis of a publicly released benchmark. The manuscript reports aggregate results, makes no re-identification attempts, and uses privacy-

preserving pixelated representative crops where visual examples are needed.

Thank you for your time and consideration.

Sincerely,

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